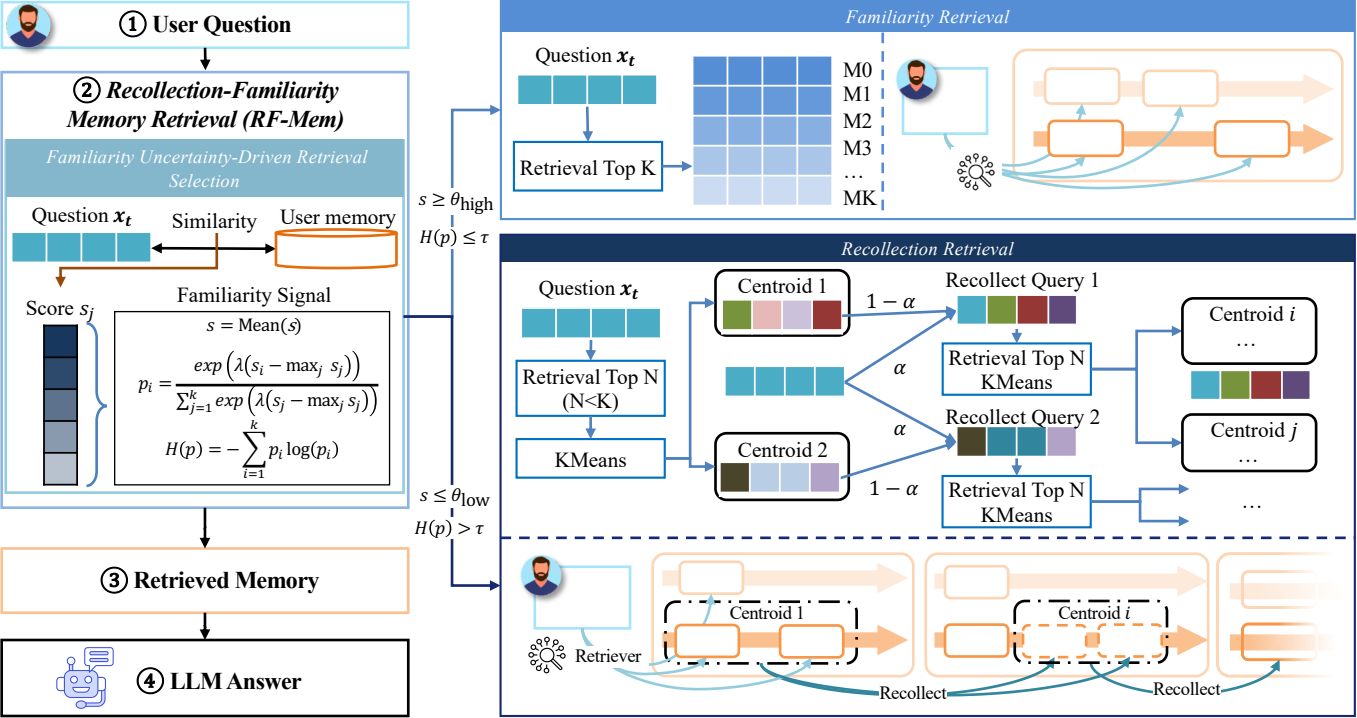
① User Question
② Recollection-Familiarity Memory Retrieval (RF-Mem)
Familiarity Uncertainty-Driven Retrieval Selection
Question x_t
Similarity
User memory
Score s_j
Familiarity Signal
$s = \text{Mean}(s)$
$p_i = \frac{exp\left(\lambda(s_i - \max_j s_j)\right)}{\sum_{j=1}^{k} exp\left(\lambda(s_j - \max_j s_j)\right)}$
$H(p) = -\sum_{i=1}^{k} p_i \log(p_i)$
③ Retrieved Memory
④ LLM Answer
$s \geq \theta_{\text{high}}$
$H(p) \leq \tau$
$s \leq \theta_{\text{low}}$
$H(p) > \tau$
Familiarity Retrieval
Question x_t
Retrieval Top K
M0
M1
M2
M3
…
MK
Recollection Retrieval
Question x_t
Retrieval Top N (N<K)
KMeans
Centroid 1
Centroid 2
$1-\alpha$
α
α
$1-\alpha$
Recollect Query 1
Recollect Query 2
Retrieval Top N KMeans
Retrieval Top N KMeans
Centroid i
…
Centroid j
…
…
Retriever
Centroid 1
Centroid i
Recollect
Recollect